Janu Kumar (J.K.) Sah

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Research Interests

Thermal and fluid sciences · Fluid-structure interaction · Bio-inspired flow sensing · Computational mechanics · Sustainable energy and materials

Education

Ph.D., Mechanical Engineering Jan 2026 – Present
Rochester Institute of Technology, Kate Gleason College of Engineering *Rochester, NY*

- **GPA:** 4.0/4.0 (to date)
- **Research project:** *From Nature to Prototype: Development of a Bio-Inspired Seal Whisker Sensor Array* (Supervisor: Dr. Qian Xue)
- **Coursework:** Computational Methods in Fluid Dynamics, Continuum Mechanics, Advanced Engineering Mathematics, Engineering Analysis, Interdisciplinary Research Methods, Translating Discovery into Practice, Doctoral Seminar

B.E., Mechanical Engineering Nov 2018 – Apr 2023
Pulchowk Campus, Institute of Engineering, Tribhuvan University *Kathmandu, Nepal*

- **Overall score:** 76.36/100; merit-based full scholarship for undergraduate study.
- **Thesis:** Study of the energy scenario and improvement opportunities of the ICT building, Pulchowk Campus (Supervisor: Dr. Shree Raj Shakya).

English proficiency: IELTS 7.0/9.0

Research Experience

Graduate Research Assistant Jan 2026 – Present
Department of Mechanical Engineering, Rochester Institute of Technology *Rochester, NY*

- Conducted towing-tank experiments with bio-inspired seal-whisker sensor arrays for underwater wake sensing.
- Debugged multi-channel sensor electronics and developed embedded firmware for data collection.
- Assembled and tested an underwater robotic vehicle as the platform for in-water sensor trials.

Research Assistant Apr 2023 – Mar 2024
Center for Energy Studies, Institute of Engineering, Tribhuvan University *Lalitpur, Nepal*

- Contributed to 25+ solar-monitoring projects and energy audits of industrial sites.
- Research focus: national energy-intensity assessment, wind-energy potential for water pumping, and industrial sustainability-index calculation.

Professional Experience

Mechanical Design Engineer Apr 2024 – Nov 2025
Ozaki Design Office *Niihama City, Japan*

- Performed CAD design, 3D modeling, and engineering simulation; conducted industrial site visits to study and analyze mechanical components.

Intern Apr – May 2022
Alternative Energy Promotion Center *Kathmandu, Nepal*

- Gained hands-on experience in energy auditing and management, renewable energy, and energy policy, regulation, and outreach.

Publications

Peer-Reviewed Journal Articles

1. **Sah, J. K.** (2026). When is carbonation-cured concrete actually cleaner? A systematic review, a net-benefit condition framework, and a proposed minimum reporting set for CO_2 mineralization in cement-based products. *Journal of Cleaner Production*, 579, 149605. doi:10.1016/j.jclepro.2026.149605
2. Shakya, S. R., **Sah, J. K.**, Bhujel, P., Aryal, H., Bajracharya, T. R., Bhandari, R., Subedi, S., & Chaulagain, B. (2024). Energy intensity improvement potential of the commercial buildings in developing countries: A case study of Nepal. *Journal of Innovations in Engineering Education*, 7(1), 84–95.
3. Shakya, S. R., Subedi, S., Bajracharya, T. R., Shrestha, S. L., **Sah, J. K.**, & Chaulagain, B. (2025). Sustainability Index for a Manufacturing Industry: A Case Study of Cement Industry in Nepal. *Journal of Science and Engineering*, 12(1), 37–48.
4. Shakya, S. R., Chaulagain, B., Poudel, S., Subedi, T. R., Poudel, P. S., Subedi, S., & **Sah, J. K.** (2025). Computation and Experimental Analysis of Helical Bach-Type Savonius Wind Turbine for Water Pumping Application. *Journal of the Institute of Engineering*, 18(1), 94–101.

5. Shah, M. P., & **Sah, J. K.** (2023). Design Optimization of Fins of a Sounding Rocket for Maximum Lift-to-Drag Ratio and Minimum Radar Cross-Section Area Using ANSYS. *International Journal on Engineering Technology*, 1(1), 53–61.

Manuscripts Under Revision

1. **Sah, J. K.** Do neural operators beat classical solvers for linear elasticity? A strong-baseline, full-cost benchmark. *Computational Mechanics* (Springer), under revision.

Peer-Reviewed Conference Proceedings

1. **Sah, J. K.**, Ghimire, I., Tamang, B., & Adhikari, N. (2025). CFX Analysis to Study the Effects of Blade Exit Angle on Performance of Centrifugal Pumps. In *Vibration Engineering and Technology of Machinery, Vol. I (VETOMAC 2022)*, Lecture Notes in Mechanical Engineering. Springer, Singapore.
2. **Sah, J. K.**, & Paudel, A. (2023). Cost Analysis of Owning Four-Wheel Electric Vehicles in Nepal. *International Conference on Engineering & Technology*, Vol. 5, KEC, 112–116.

Selected Projects

- **Industrial Energy Audit, Dairy Development Corporation, Kathmandu.** Proposed an integrated recovery system: 138 KVA/month demand reduction via power-factor correction (0.67→0.98) and 15% boiler-fuel savings through insulated cladding (2-month payback).
- **EV Cost-Competitiveness Model (Kathmandu Valley).** MATLAB lifetime-cost model showing 40% lower cost for converted EVs vs. new ICE vehicles at 10,000 km/yr.

Technical Skills

- **Programming & Data:** Python, MATLAB; signal processing (FFT, spectral analysis); machine learning (neural operators).
- **Simulation & Modeling:** CFD and fluid-structure interaction (ANSYS CFX/Fluent, immersed-boundary methods); finite element method; reduced-order models.
- **Experiments & Instrumentation:** towing-tank testing, strain-gauge sensors, SPI sensor interfacing, ESP32 firmware, PCB debugging, oscilloscope / multimeter benchtop testing.
- **Thermal & Energy:** building and industrial energy audits; EnergyPlus, LEAP, PVsyst.
- **CAD:** SOLIDWORKS, AutoCAD, SketchUp (3D modeling and drafting).
- **General:** LaTeX, Git; experimental design, data analysis, and technical writing.

Talks and Presentations

Oral presentation | 2026 RIT Graduate Showcase, Rochester, NY — Apr 2026
From Nature to Prototype: Development of a Bio-Inspired Seal Whisker Sensor Array

Exhibit | Imagine RIT, Rochester, NY — Apr 2026
From Nature to Prototype: Seal Whisker Sensors

Poster | VETOMAC 2022, Tribhuvan University, Nepal — Dec 2022
Effects of Blade Exit Angle on the Performance of a Centrifugal Pump

Honors and Awards

RIT Ph.D. Merit Scholarship / Graduate Assistantship — Jan 2026 – Present

Merit-Based Full Scholarship for Undergraduate Study — Nov 2018 – Apr 2023

Professional Service, Leadership & Outreach

- **Reviewer**, *International Journal of Concrete Structures and Materials* (Springer).
- **Reviewer**, *Journal of Building Engineering* (Elsevier).
- **President**, Saraswati Pooja Committee, NTBNS, IOE Pulchowk Campus (2022–2023).
- **Exhibition Coordinator**, 13th National Mechanical and Aerospace Engineering Exhibition, SOMAES (2022–2023).